3.3 Planning

Students will be expected to plan an experiment set by Pearson, although they will not be expected to carry it out.

Candidates will be assessed on their ability to:

Plan an experiment	• identify the apparatus required • the range and resolution of measuring instruments including Vernier calipers (0.1mm) and micrometer screw gauge (0.01mm) • discuss calibration of instruments, e.g. whether a meter reads zero before measurements are made • describe how to measure relevant variables using the most appropriate instrument and correct measuring techniques • identify and state how to control all other relevant variables to make it a fair test • discuss whether repeat readings are appropriate • identify health and safety issues and discuss how these may be dealt with • discuss how the data collected will be used • identify possible sources of uncertainty and/or systematic error and explain how these may be reduced or eliminated • comment on the implications of physics (e.g. benefits/risks) and on its context (e.g. social/environmental/historical).